

Shopee Taiwan: Providing fun and seamless online shopping experiences with cloud infrastructure

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About Shopee Taiwan

To enhance its IT maintenance efficiency, Shopee Taiwan leverages Google Cloud for infrastructure consolidation and high connectivity.

Shopee Taiwan is the Taiwanese branch of Shopee, a leading ecommerce platform in Taiwan and Southeast Asia. Founded in 2015, Shopee provides tailored services for the region featuring an easy, secure, and fast online shopping experience through strong payment and logistical support. Besides traditional ecommerce services, Shopee also offers games and social media features on its online shopping application.

Industries: Retail & Consumer Goods

Google Cloud results

- Simplifies maintenance work with the pre-configuration features of GKE and Cloud SQL
- Converts scanned images to editable text formats with more than 99% accuracy through Vision AI
- Shortens data inquiry time to identify in-app errors in five minutes with BigQuery
- Reduces app response time to five milliseconds

Location: Taiwan

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Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)
Vision AI (<https://cloud.google.com/vision/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Vision AI (<https://cloud.google.com/vision/>)

with Cloud CDN

Most people are familiar with online shopping, but what Shopee Taiwan (<https://shopee.tw/>) offers is more than just buying products on its website and application. As the Taiwanese branch of Shopee, a leading ecommerce platform in Southeast Asia and Taiwan founded in 2015, Shopee Taiwan provides customers with an easy, secure, and fast online shopping experience, and has released several games as well as social media features on its shopping app. It also launched Shopee University, which is a learning platform for ecommerce sellers.

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—Chris Cheng, Engineer,
Tech Team, Shopee
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“We want to offer online consumers a one-stop platform with a wide variety of products and diverse entertainment features,” says Ashley Lai, senior product manager at Shopee Taiwan. “Besides an easy and fast shopping experience, our users can also enjoy social interaction and excellent media content on our shopping app.”

Shopee Taiwan previously deployed its services on on-premises servers. But as the range of its services expanded, and the flow of its application increased, its IT team realized that in order to ensure service quality, it needed an infrastructure that is more easy to scale and can be maintained more efficiently. For example, Shopee Taiwan’s games can attract a lot of users during big promotion campaigns, which would be challenging to handle with a limited amount of on-premises computing resources.

After comparing different public cloud offerings, Shopee Taiwan migrated its services in early 2019 to Google Cloud because the latter has an office and excellent partners in Taiwan to provide immediate and face-to-face technical support, as well as a local data center that supports 2X faster connection speed for users in Taiwan.

“Besides outstanding products that can help facilitate our work, we’re also looking for a cloud provider that is able to offer prompt technical support and the fastest connection speed for our users in Taiwan. Google Cloud is the only one that ticks all the boxes,” recalls Chris Cheng, engineer at Shopee Taiwan.

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Enhancing maintenance efficiency and scalability with GKE and Cloud SQL

With the help of the Google Cloud team, who provided advice for infrastructure design and product choice, Shopee Taiwan smoothly completed the migration in three months. Currently, the company runs the Kubernetes containers of its delivery system, in-app games, and learning platform for ecommerce sellers on [Google Kubernetes Engine](#) (/kubernetes-engine) (GKE) and stores all its business data on [Cloud SQL](#) (/sql).

Cheng says that the pre-configuration feature of GKE and Cloud SQL has significantly simplified the maintenance work for Shopee Taiwan’s IT team. Before the migration, its engineers had to set up all the parameters one by one, which took a great amount of time. With GKE and Cloud SQL, the team can apply the pre-configured operation environments that fit its needs with one click. GKE also comes with many monitoring plug-in tools that greatly reduce the maintenance effort required.

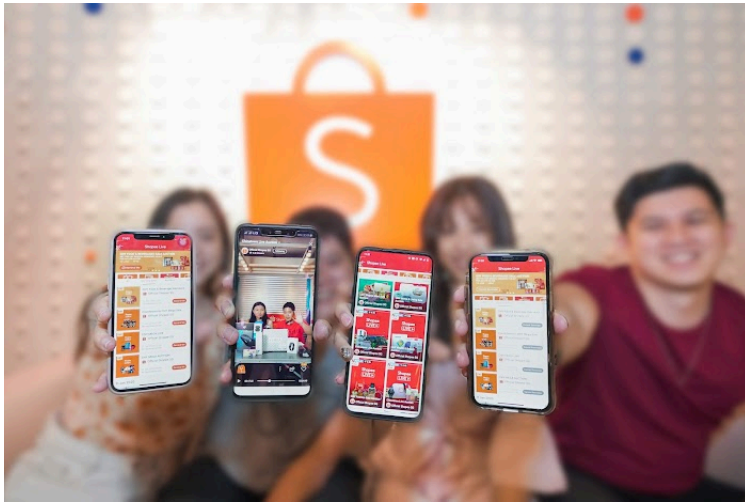
“We have benefited a lot from the high-maintenance efficiency supported by GKE and Cloud SQL,” notes Cheng. “Now, we are able to take care of a maintenance workload that is 10X more than before.”

The high scalability feature of GKE has also enabled Shopee Taiwan to handle instant flow surges by quickly launching new virtual machines. During sale promotions, the flow from Shopee Taiwan’s in-app games to its marketplace can sometimes increase by up to 30X in a short period of time. With GKE, the company is able to ensure a great user experience, regardless of the level of flow surge.

Facilitating internal administrative work with Vision AI

Shopee Taiwan needs to import paper documents into its digital system every month. To enhance input efficiency, the company adopted Cloud Vision (/vision) for its high image-to-text recognition accuracy.

“The optical character recognition software that we previously used could only accurately process the images with good quality, but Vision AI can perfectly convert images into text despite the image quality,” says Cheng. “The overall conversion error rate of Vision AI is lower than 1%, which has significantly facilitated our administrative work.”



Troubleshooting efficiently with BigQuery

When an error occurs in Shopee in-app games, its engineers need to consult the log data of the games to identify the cause. To maximize the troubleshooting efficiency, the team imports the log data from Cloud SQL to BigQuery ([/bigquery](#)) for faster inquiry speed. According to Cheng, an inquiry that normally takes 10–15 minutes on Cloud SQL requires only 30 seconds on BigQuery, which has greatly reduced the time needed to clear errors and helped deliver better gaming experiences.

“BigQuery is a very powerful tool for data inquiry. By storing the log data of our games on BigQuery, our engineers are able to quickly go through the data and pinpoint the cause of an error in five minutes,” he adds. “This way, users can fully enjoy our games with bugs being cleared almost instantly.”

Supporting fast content delivery across borders with Cloud CDN

To optimize the loading speed of its online shopping app, Shopee Taiwan leverages Cloud CDN (/cdn) to deliver app content and stores its CDN configuration data on Cloud Storage (/storage). By caching content in proxy servers that are located closer to end users globally, Cloud CDN not only supports faster content delivery speed, but also helps Shopee Taiwan guarantee consistently high connectivity when it releases its games outside Taiwan. The company has made its games available in Thailand, Vietnam, and Singapore, with plans to expand further.

“The high connectivity supported by Cloud CDN through its global network enables us to provide a glitchless user experience and expand to foreign markets easily,” explains Cheng. “Thanks to Cloud CDN, the response time of our app is reduced to five milliseconds, and our users in different countries can enjoy the same loading speed.”

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*—Chris Cheng, Engineer,
Tech Team, Shopee
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Continuously optimizing online shopping experiences

Shopee Taiwan is always looking for ways to optimize the online shopping experiences that it offers. Moving forward, the company plans to use Cloud Monitoring (/monitoring) to further increase the monitoring efficiency of its IT team and enhance the stability of its services by preventing errors from happening based on monitoring information. It will

also employ Cloud Tasks (/tasks) to reduce request latency on its app.

Cheng says, “Google Cloud has enabled us to create fun and glitchless online shopping experiences by providing a cloud infrastructure that supports high performance and is easy to manage. Seeing this, we’re confident that we can continue improving our user experience while introducing new features in the future.”